United International **University** (**UIU**)

Dept. of Computer Science and Engineering (CSE)

Final Exam Year: **2026** Semester: **Spring**

Course: **CSE 3711** Title: **Computer Networks (Section – ALL)**

Marks: **40** Time: **2 Hours**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **Three (03)** questions. Answer **all of them**. All questions are of values indicated on the right-hand margin.

IMPORTANT NOTE: Candidates are required to answer the questions in **sequential order** (Q1, Q2 and Q3).

Q1. a) **List at least two (2)** differences between **TCP** and **UDP header formats**. **[2]**

b) Suppose, **Host A**, with **IP address 172.16.10.111** and **source port 7000**, sends an **HTTP request** to **google.com**, a web server that gives secured **HTTP service** using IP address **142.251.211.78** and port number **443**, and **Host B**, with **IP address 192.168.11.110** sends 2 (Two) requests: an **HTTP request** with **source port 8000** to the same web server (IP address **142.251.211.78** and port number **443)**, and an **SMTP request** with **source port 9000** to **gmail.com**, an email server that gives web-based **email service** using IP address **142.251.16.109** and port number **587**. **Describe** how **multiplexing** and **demultiplexing** occur at the Transport layer for the above scenario. **[3]**

c) What is the **advantage** of **fast retransmission** (**3-dup**) over the **timeout** mechanism for retransmission in the original TCP? **Explain**. **[2]**

d) **Host P** initiates a TCP connection with **Host Q** using the **3-way handshake** process consisting of Step 1, Step 2, and Step 3.

i. In **which steps** is the **ACK flag** set to **1**? **[1]**

ii. If **Host P** sends an initial sequence number (**Seq #**) of **4100** in **Step 1**, what will be the acknowledgment number (**ACK #**) sent by **Host Q** in **Step 2**? **[2]**

e) **Host A** and **B** are communicating over a TCP connection, and **Host B** has already received from **Host A** all bytes up to **byte # 1000**. Suppose **Host A** then sends **two segments** to **Host B** back-to-back. The first and second segments contain **200** and **500 bytes of data**, respectively. In the **first segment**, the **sequence number** is **1001**, the **source port number** is **5555**, and the **destination port number** is **25**. **Host B** sends an acknowledgement whenever it receives a segment from **Host A**.

i. If the **second segment** arrives before the **first segment**, in the acknowledgment of the first arriving segment, what is the **acknowledgment number, source port number,** and **destination port number**? **[2]**

ii. Suppose the **two segments** sent by A arrive in order at B. The **first acknowledgement is lost** and the second acknowledgement arrives **after** the first timeout interval. **Draw a timing diagram**. Describe **what will happen** after the **first timeout interval**? Retransmission, or new segment? **[3]**

Q2. a) **Illustrate** the difference between **Routing** and **forwardin**g in network layer with an **example**. **[2]**

b) Consider the network shown in the following diagram (**Figure 1**) as a graph $G = (N, E)$, where $\boldsymbol{N}$ is the set of routers and $\boldsymbol{E}$ is the set of links, use **Dijkstra's link-state routing algorithm** to compute the least cost path from **node B** to all other nodes and show the resulting **forwarding table** for **B**. **Show all calculations** to get full credit. **[5]**

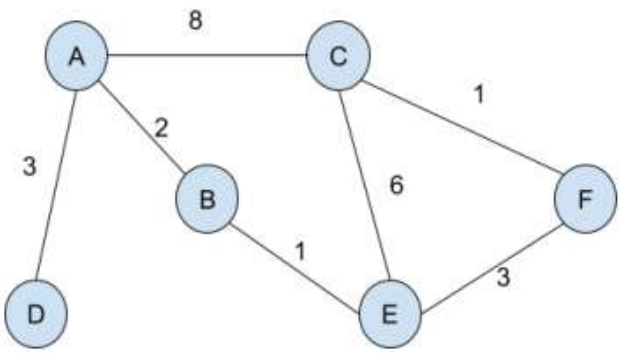
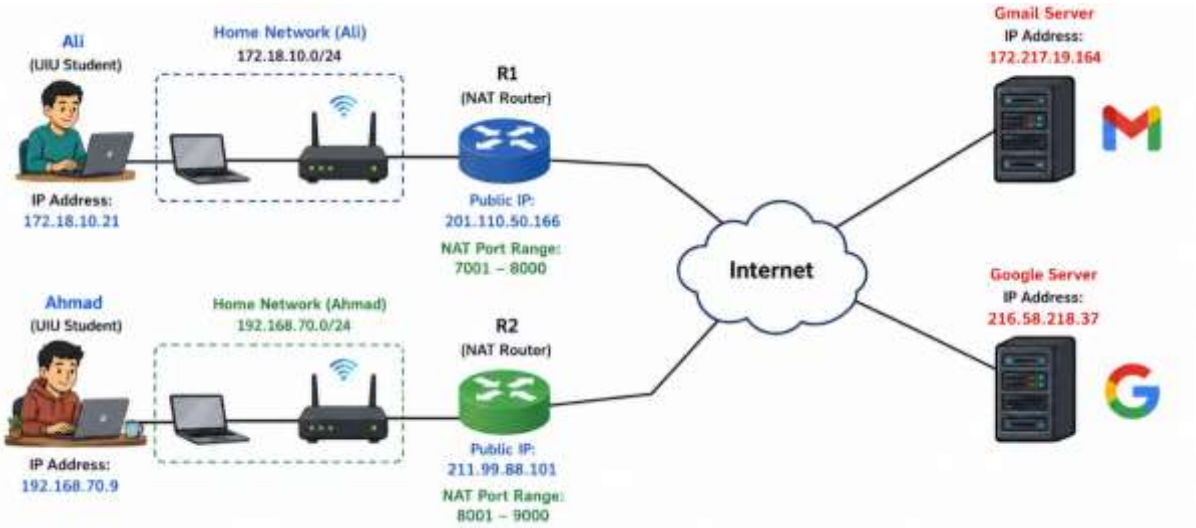


Figure 1: *Graph for Q.2(b)*

c) A datagram of **4,500 bytes** arrives at a router that should be forwarded to a link with an **MTU of 1,500 bytes**. Suppose the original datagram is stamped with **an identification number of 888**. Assume that the size of the **IP header is 30 bytes**. With a diagram show different fragments including the length, ID, ***fragflag*** and offset values. **[4]**

Q2. d) Consider the following diagram. **Ali** and **Ahmad** are two of the UIU students browsing Internet from home behind their NAT routers, **R1** and **R2** respectively. **Ali**'s IP address is **172.18.10.21** and **Ahmad**'s IP address is **192.168.70.9**. **Router R1** is using public IP **201.110.50.166** and port numbers from **7001 to 8000** for NAT. **Router R2** is using public IP **211.99.88.101** and port numbers **8001 to 9000** for NAT. Also, assume that **Gmail Server's** IP address is **172.217.19.164** and **Google Server**'s IP address is **216.58.218.37**.



i. Suppose, **Ali** is accessing **Gmail Server** from his PC using client **port # 5555**. What will be the **source** & **destination IP address** and **port numbers** in the packet arriving at the **router R1** from **Gmail Server**. **[2]**

ii. Now, suppose both **Ali** and **Ahmad** are **checking E-mail** (using server port # 25 and client port # 5500) and **searching some materials in Google** (using server port # 443 and client port # 6600) simultaneously. Show the **corresponding entries** in **NAT table** for both router R1 & R2. **[3]**

Q.3 a) Briefly describe **any 4 (four) services** provided by data link layer. **[2]**

b) Consider the figure below (**Figure 2**). The **IP and MAC addresses** are shown for nodes **A, B, C** and **D**, as well as for the **router's interfaces**. Consider **an IP datagram** being sent from node **C** to node **A**. Assume, **ARP tables** are **empty**.

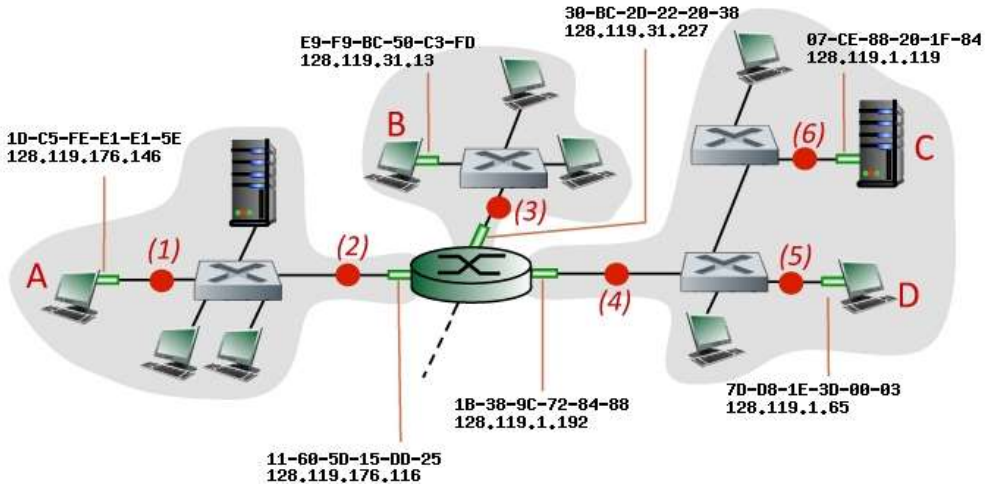


Figure 2: *Network Topology for Q.3(b)*

i. What is **Address Resolution Protocol (ARP)**? How **default gateway** plays an important role in ARP? **[2]**

ii. If **host C** runs the **ARP protocol**, what would be the **source/destination IP addresses** and **source/destination MAC addresses** of the **ARP Request packet**? **Justify** your answer. **[2]**

iii. **Which device** will send **ARP reply** to the ARP request sent by **C**? **[1]**

iv. Give the **source and destination MAC addresses**, as well as the **source and destination IP addresses** encapsulated within the **Ethernet data frame** from C to A at points ***(6),(4),(2)*** and ***(1)*** in the figure. **[2]**

END OF THE QUESTION